UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

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|---|---|---|
| COURSE | : | INTERACTIVE MULTIMEDIA |
| COURSE CODE | : | CSC413 |
| EXAMINATION | : | NOV 2026 (MOCK) |
| TIME | : | 3 HOURS |

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

*This marking scheme consists of 6 printed pages*

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## PART A - ANSWER KEY

| Q | Ans | Statement | Explanation |
|---|---|---|---|
| A1 | TRUE | Authoring tools help developers organise and edit multimedia elements. | Authoring tools assemble media. |
| A2 | FALSE | 3D animation does not use modelling. | 3D animation relies on models. |
| A3 | TRUE | A storyboard is a visual plan of a multimedia sequence. | Storyboards show the planned flow. |
| A4 | TRUE | Hertz measures sound frequency. | Hertz is cycles per second. |
| A5 | FALSE | A vector image becomes pixelated when enlarged. | Vector images scale without pixelation. |
| A6 | TRUE | MPEG is a common family of video compression formats. | MPEG defines video compression standards. |
| A7 | FALSE | Anti-aliasing makes the edges of text look more jagged. | Anti-aliasing smooths jagged edges. |
| A8 | TRUE | CMYK is used for printed output. | CMYK is a subtractive print model. |
| A9 | FALSE | MIDI files store digitized audio samples. | MIDI stores musical instructions, not samples. |
| A10 | FALSE | A plotter is an input device. | A plotter produces output. |
| A11 | TRUE | Hypermedia combines multimedia with hyperlinks. | Hypermedia adds linked navigation to multimedia. |
| A12 | TRUE | A web browser displays hypertext documents. | Browsers render web pages. |
| A13 | TRUE | USB is a plug-and-play connection standard. | USB devices are recognised automatically. |
| A14 | TRUE | RGB is an additive colour model. | RGB adds red, green and blue light. |
| A15 | FALSE | Interlacing displays every line of a video frame at the same time. | Interlacing draws alternate lines. |

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## PART B - ANSWER KEY

| Q | Ans | Term | Definition |
|---|---|---|---|
| B1 | J | Multimedia | integration of text, graphics, audio, video and animation |
| B2 | M | Virtual reality | simulated environment |
| B3 | A | Frame rate | frames displayed per second |
| B4 | I | Bitmap | image made of pixels |
| B5 | O | Storyboard | visual sequence plan |
| B6 | L | Keyframe | important frame in an animation |
| B7 | C | Browser | software that displays web pages |
| B8 | G | Scanner | input device that digitises documents |
| B9 | D | Linear multimedia | presentation without user control of sequence |
| B10 | F | Vector | image made of mathematical objects |
| B11 | E | Hypermedia | linked multimedia elements |
| B12 | B | HTML | language for web structure |
| B13 | H | Copyright | exclusive rights of a creator |
| B14 | N | Tweening | generating in-between frames |
| B15 | K | Card-based tool | authoring tool organised like pages or cards |

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## PART C - ANSWER KEY

| Q | Ans | Correct answer |
|---|---|---|
| C1 | C | A low-fidelity layout sketch |
| C2 | A | Card-based |
| C3 | B | Links to other content |
| C4 | C | WAV |
| C5 | C | Compiler |
| C6 | C | Structure web content |
| C7 | B | Text, graphics, audio, video and animation |
| C8 | A | One shape into another |
| C9 | B | Temporary working memory |
| C10 | C | Frames per second |
| C11 | B | PNG |
| C12 | A | In-between frames |
| C13 | B | An early model of a product |
| C14 | A | Draws alternate lines of a frame |
| C15 | D | It combines visuals and sound |
| C16 | C | Screen display |
| C17 | A | Scanner |
| C18 | A | Non-linear |
| C19 | D | Analog sound to digital values |
| C20 | B | Modelling and rendering |
| C21 | B | Small strokes at character ends |
| C22 | C | Important frames in an animation |
| C23 | B | Messages with a beginning and end |
| C24 | B | CMYK |
| C25 | C | Frequency |
| C26 | C | A cinema presentation |
| C27 | B | Loudness |
| C28 | A | Another resource |
| C29 | A | Merging of media and computing |
| C30 | D | Automatic device recognition |

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# PART D - WORKED ANSWERS

## QUESTION 1

*Topic: text*

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

**a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.**

Sans serif is cleaner at screen sizes; serif aids reading in print.

*(3 marks)*

**b) Describe TWO roles of hypertext in multimedia.**

It links related content and supports non-linear navigation.

*(3 marks)*

**c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.**

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

*(4 marks)*

## QUESTION 2

*Topic: authoring interactivity*

A company wants an interactive product demonstration that runs on a kiosk and on the web.

**a) State TWO features to consider when choosing authoring software.**

Interactivity support and export formats are important features.

*(2 marks)*

**b) Explain the interactivity level suitable for the demonstration.**

A medium level with menus and buttons lets users explore without overwhelming them.

*(3 marks)*

**c) Explain TWO authoring tool categories appropriate for this project.**

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

*(5 marks)*

## QUESTION 3

*Topic: web hypermedia*

A museum wants a website that presents its collection and lets visitors explore related exhibits.

**a) State TWO differences between linear and non-linear navigation.**

Linear follows a fixed sequence; non-linear lets users choose their own path.

*(2 marks)*

**b) Describe TWO roles of hyperlinks in the website.**

They connect related exhibits and let users move between pages.

*(4 marks)*

**c) Explain how streaming and compression affect multimedia delivery on the web.**

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

*(4 marks)*

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## QUESTION 4

*Topic: design project*

A shopping complex requires a kiosk that gives customers location and retailer information.

**a) State TWO important project team roles for completing the kiosk.**

A project manager and an interface designer are important roles.

*(3 marks)*

**b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.**

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

*(3 marks)*

**c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.**

Portability and constant connectivity make mobile devices convenient for games and shopping.

*(4 marks)*

*END OF MARKING SCHEME*

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